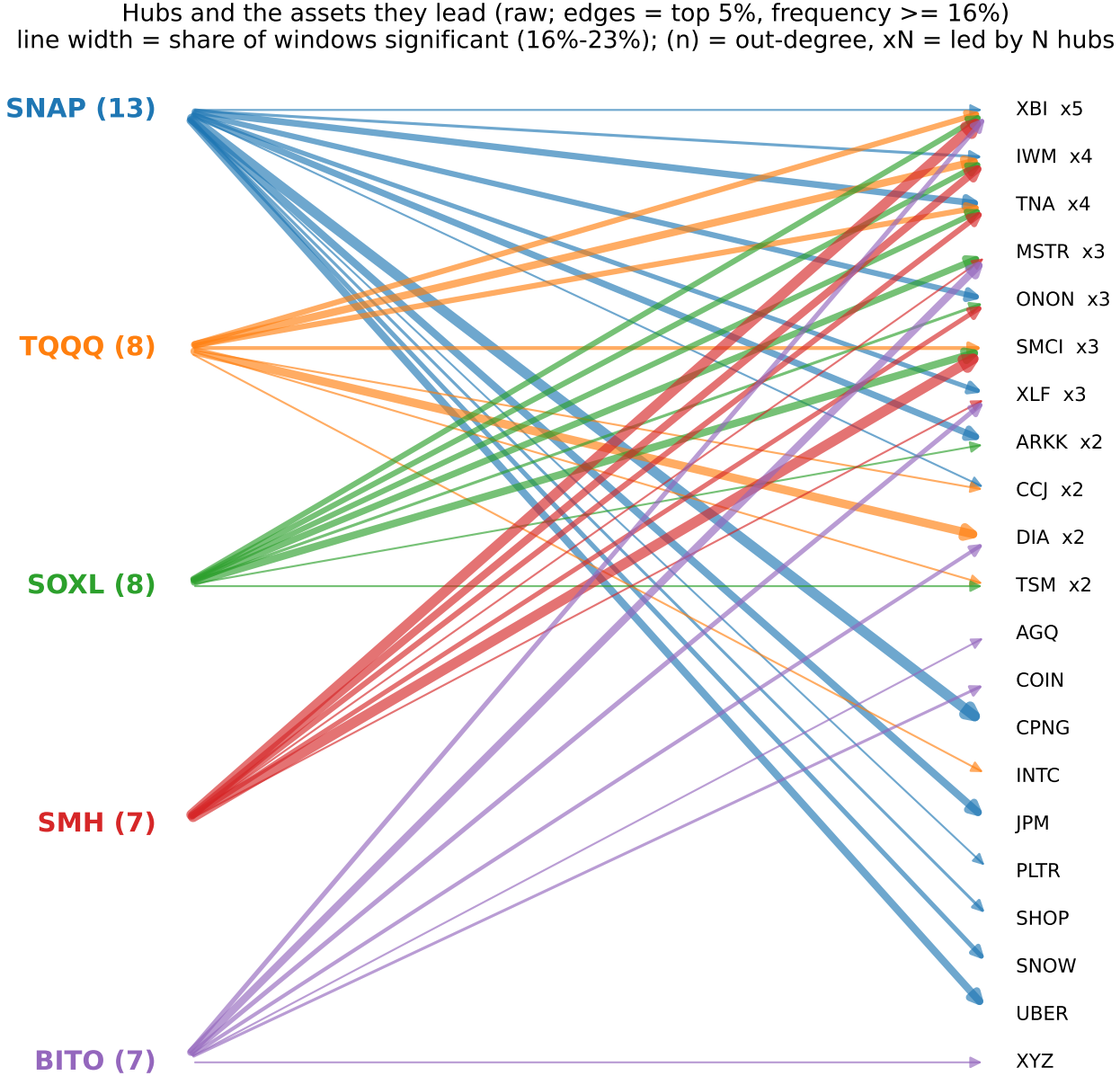
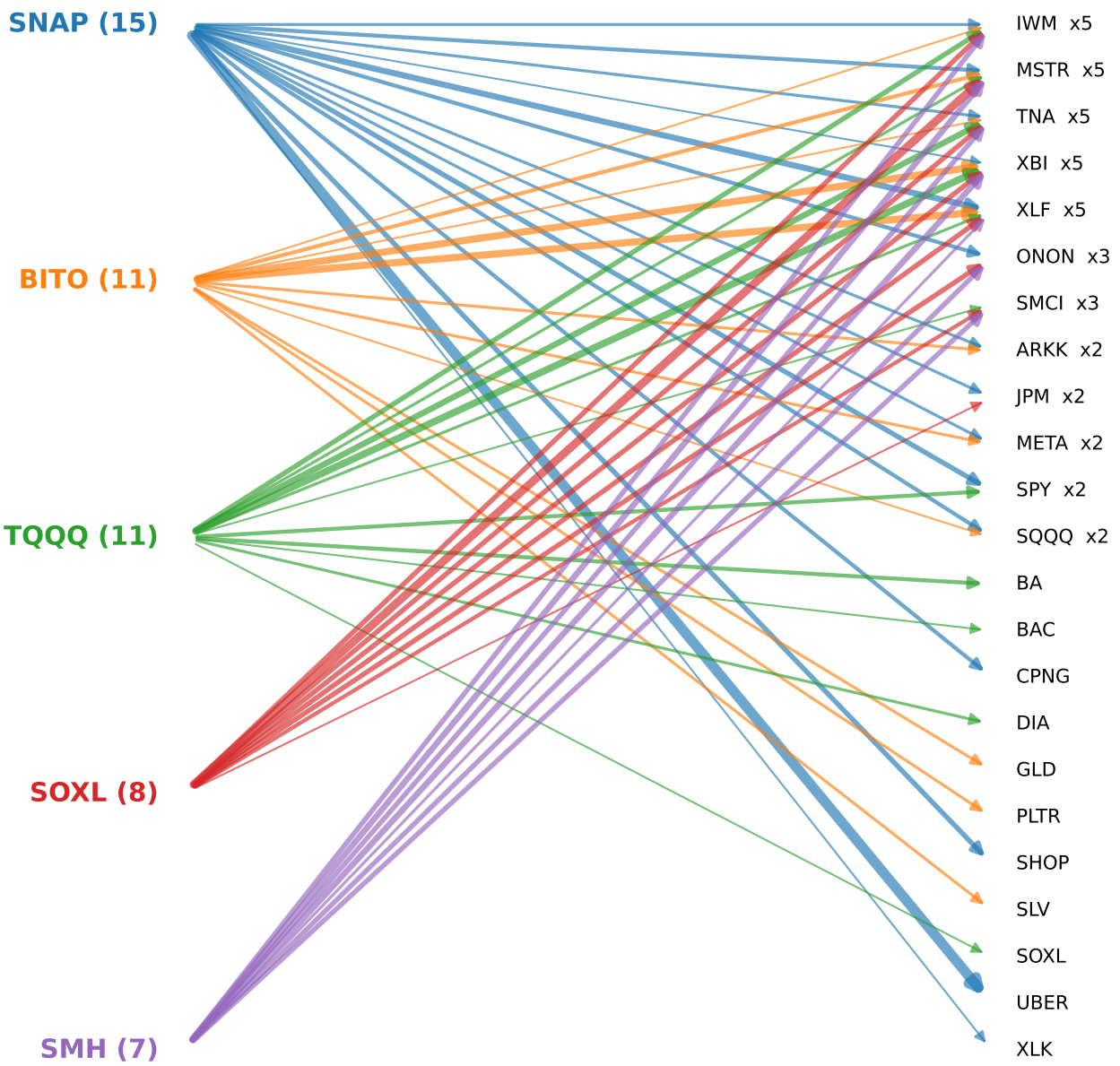
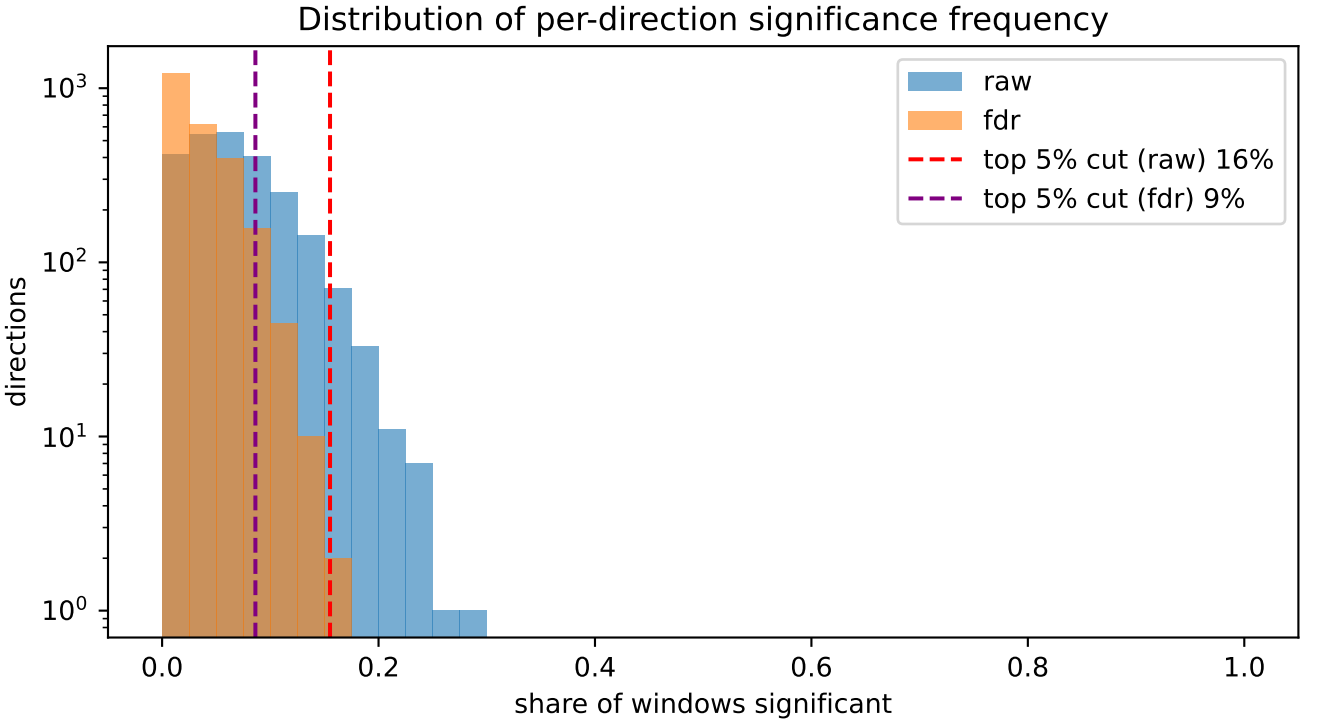
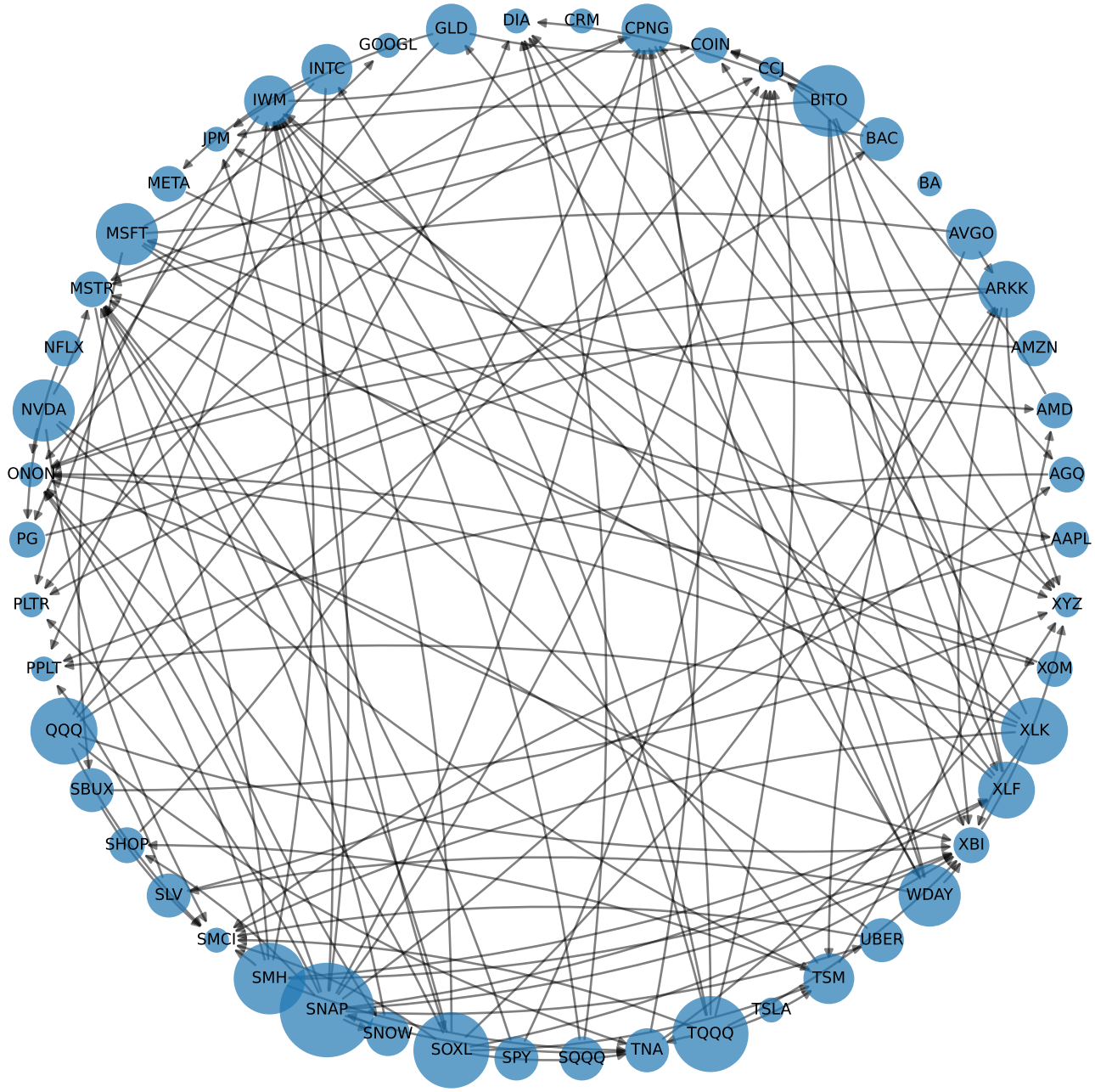
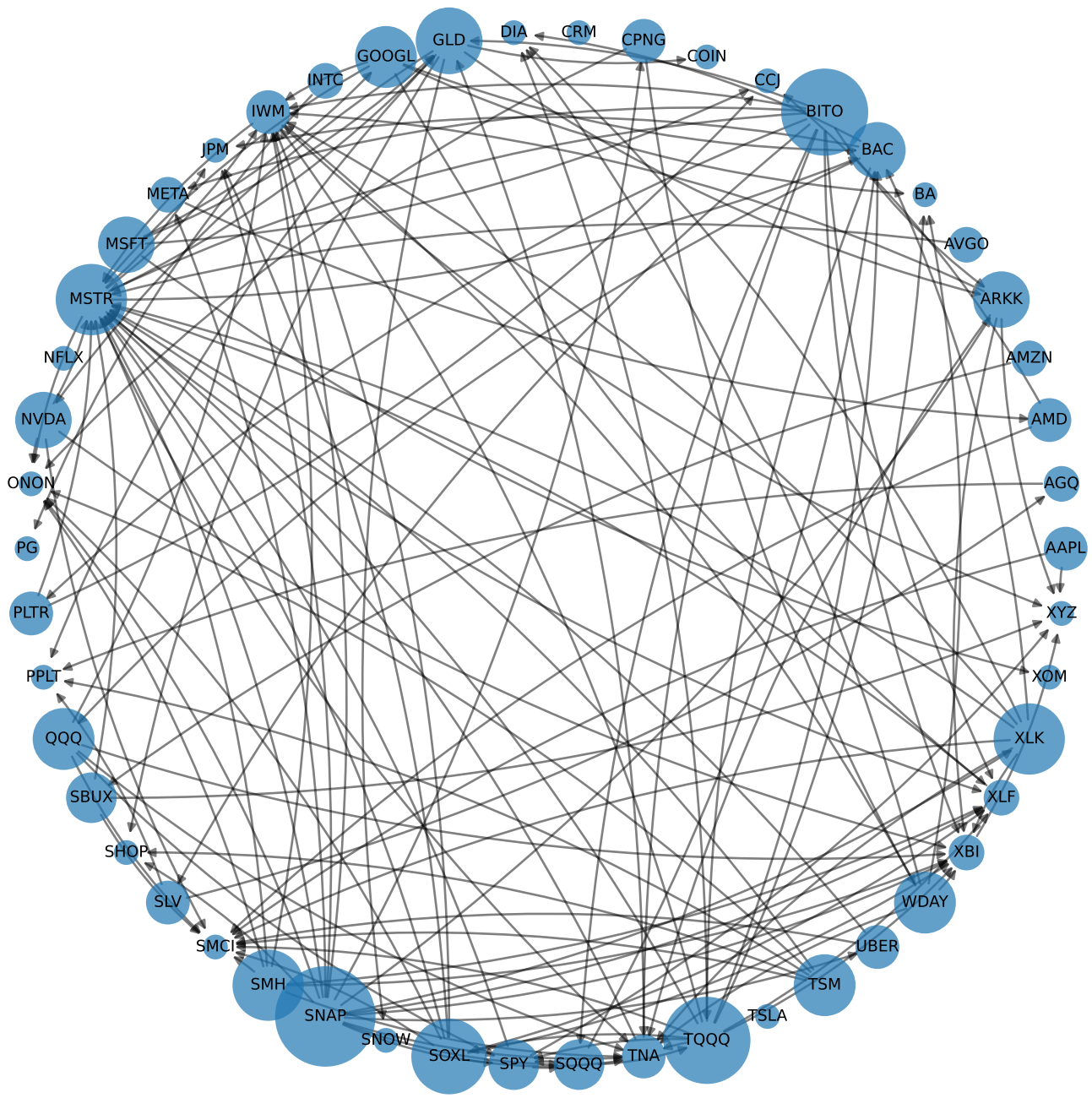
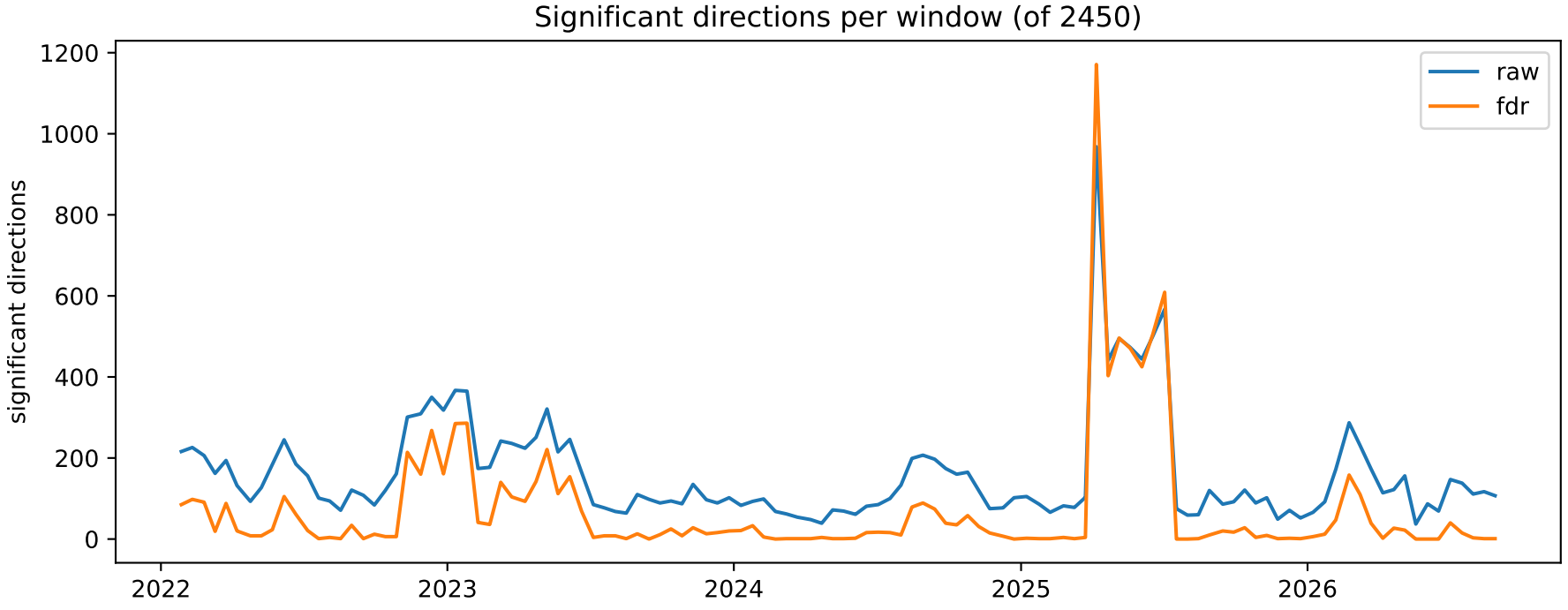

Hubs and the assets they lead (raw; edges = top 5%, frequency >= 16%)
line width = share of windows significant (16%-23%); (n) = out-degree, xN = led by N hubs
SNAP (13)
TQQQ (8)
SOXL (8)
SMH (7)
BITO (7)
XBI x5
IWM x4
TNA x4
MSTR x3
ONON x3
SMCI x3
XLF x3
ARKK x2
CCJ x2
DIA x2
TSM x2
AGQ
COIN
CPNG
INTC
JPM
PLTR
SHOP
SNOW
UBER
XYZ


Hubs and the assets they lead (fdr; edges = top 5%, frequency >= 9%)
line width = share of windows significant (9%-16%); (n) = out-degree, xN = led by N hubs
SNAP (15)
BITO (11)
TQQQ (11)
SOXL (8)
SMH (7)
IWM x5
MSTR x5
TNA x5
XBI x5
XLF x5
ONON x3
SMCI x3
ARKK x2
JPM x2
META x2
SPY x2
SQQQ x2
BA
BAC
CPNG
DIA
GLD
PLTR
SHOP
SLV
SOXL
UBER
XLK


Distribution of per-direction significance frequency
raw
fdr
top 5% cut (raw) 16%
top 5% cut (fdr) 9%
directions
10^3
10^2
10^1
10^0
0.0
0.2
0.4
0.6
0.8
1.0
share of windows significant


Causality network (raw, top 5%, frequency >= 16%): 124 edges; node size = out-degree
GLD
DIA
CRM
CPNG
GOOGL
COIN
INTC
CCJ
IWM
BITO
JPM
BAC
META
BA
MSFT
AVGO
MSTR
ARKK
NFLX
AMZN
NVDA
AMD
ONON
AGQ
PG
AAPL
PLTR
XYZ
PPLT
XOM
QQQ
XLK
SBUX
XLF
SHOP
XBI
SLV
WDAY
SMCI
UBER
SMH
TSM
SNAP
TSLA
SNOW
TQQQ
SOXL
SPY
SQQQ
TNA


Causality network (fdr, top 5%, frequency >= 9%): 140 edges; node size = out-degree
GLD
DIA
CRM
CPNG
COIN
CCJ
BITO
BAC
BA
AVGO
ARKK
AMZN
AMD
AGQ
AAPL
XYZ
XOM
XLK
XLF
XBI
WDAY
UBER
TSM
TSLA
TQQQ
TNA
SQQQ
SPY
SOXL
SNOW
SNAP
SMH
SMCI
SLV
SHOP
SBUX
QQQ
PPLT
PLTR
PG
ONON
NVDA
NFLX
MSTR
MSFT
META
JPM
IWM
INTC
GOOGL


Significant directions per window (of 2450)
significant directions
1200
1000
800
600
400
200
0
2022
2023
2024
2025
2026
raw
fdr